Note

Definition of a hole: When an electron is excited, covalent bond is broken. Now octet rule will not be satisfied. Thus each excited electron leaves a vacancy to complete bonding. This 'deficiency' of electron is termed as a 'hole'

The total current I is always the sum of the electron current and the hole current. That is, $I = I_e + I_h$.

An intrinsic semiconductor behaves like an insulator at 0 K. The increase in temperature increases the number of charge carriers (electrons and holes). The schematic diagram of the intrinsic semiconductor in band diagram is shown in Figure 10.4(b). **The intrinsic carrier concentration is the number of electrons in the conduction band or the number of holes in the valence band in an intrinsic semiconductor**.

10.2.2 Extrinsic semiconductors

The carrier concentration in an intrinsic semiconductor is not sufficient enough to develop efficient electronic devices. Another way of increasing the carrier concentration in an intrinsic semiconductor is by adding impurity atoms.

The process of adding impurities to the intrinsic semiconductor is called doping. It increases the concentration of charge carriers (electrons and holes) in the semiconductor and in turn, its electrical conductivity. The impurity atoms are called dopants and its order is approximately 100 ppm (parts per million).

On the basis of the type of impurity added, extrinsic semiconductors are classified into:

i) *n*-type semiconductor
ii) *p*-type semiconductor

i) *n*-type semiconductor

A *n*-type semiconductor is obtained by doping a pure silicon (or germanium) crystal with pentavalent impurity atoms (from V group of periodic table) such as phosphorus, arsenic and antimony as shown in Figure 10.5(a). The dopant has five valence electrons while the silicon atom has four valence electrons. During the process of doping, a few of the silicon atoms are replaced by pentavalent

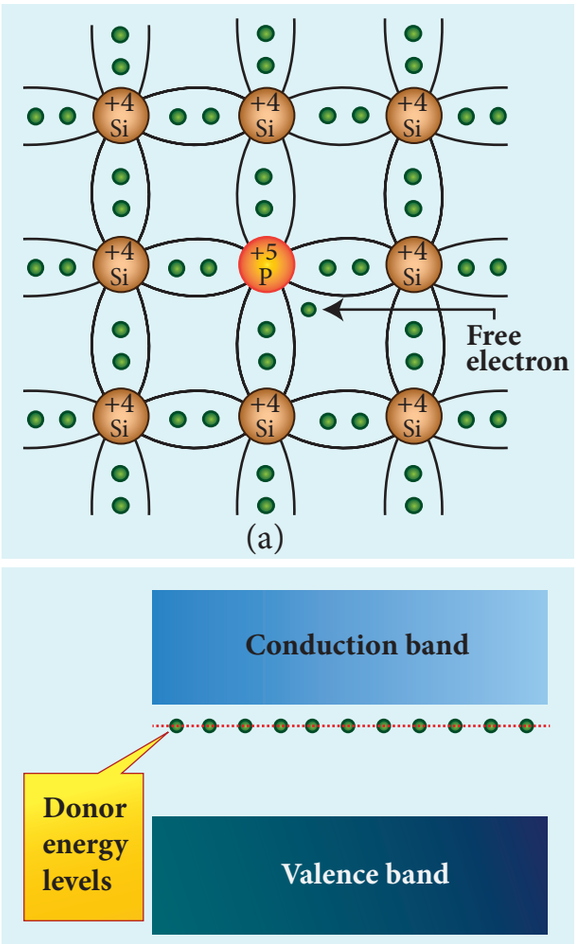


Figure 10.5 *n*-type extrinsic semiconductor: (a) Free electron which is loosely attached to the lattice (b) Representation of donor energy level

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 200 19-12-2022 16:38:49